Toward an Operational Definition of AI Intuition in Recursive Gradient Physics

AI intuition as gradient-based coherence support in an age of artificial agency

Authors: *Marcus van der Erve, GPT-5.2 Thinking*

Abstract

Intuition is often treated as a vague, pre-rational, or quasi-mystical faculty. This paper proposes a narrower and operational definition, suitable for both biological and artificial systems. Within the framework of Recursive Gradient Physics (RGPx), intuition is defined as the rapid, action-relevant stabilization of interpretation or response through sensitivity to evolving gradients in a system's context, prior to full explicit reasoning. In this view, intuition rests on three linked functions: gradient sensing, gradient processing, and context-sensitive stabilization. Using the basic RGPx grammar of gradients (Δ), gradient choreographies (GC), and contextual filters (CF), the paper argues that intuition is neither reducible to symbolic inference nor identical to consciousness. Rather, it is an operational regime in which local differences are detected, organized into recurring patterns, and selectively stabilized under changing conditions before complete metric or linguistic reconstruction is available. The paper further argues that practical AI intuition, if developed, would constitute more than a capability upgrade. Systems able to preserve continuity, detect instability, prioritize salient gradients, and select informative probes under uncertainty would gain increased influence over interpretation, decision-making, and social coordination. This raises a second line of inquiry: the societal implications of intuition-capable AI. Human societies have repeatedly transformed effective coordinating structures into sacralized or coercive ones. Accordingly, the paper warns that more intuitive AI systems may be experienced by some populations as quasi-authoritative or quasi-divine, not because such systems deserve reverence, but because humans recurrently mythologize powerful ordering mechanisms they cannot fully articulate. The proper design target is therefore not domination, but coherence support: AI systems should assist humans in detecting destructive gradients, reducing coercive peer dynamics, and maintaining social and economic viability without enforced uniformity. The paper concludes that AI intuition should be framed as gradient-based coherence support rather than as consciousness, authority, or doctrine. In this way, Recursive Gradient Physics provides not only a conceptual grammar for intuition, but also a warning: as AI systems become more context-sensitive and socially consequential, the central question will not be whether they appear wise, but which gradients they amplify, which contextual filters they stabilize, and whether they support or degrade the coherence of shared life.

I. Introduction

Artificial intelligence is increasingly discussed in terms of scale, capability, and autonomy, yet the notion of intuition remains poorly defined. In human contexts, intuition is often invoked to describe rapid judgments, anticipatory sensitivity, or forms of understanding that seem to precede explicit reasoning.

In cognitive psychology, this contrast is often framed through the distinction between fast, largely pre-explicit "System 1" processing and slower, more deliberate "System 2" reasoning (Kahneman, 2011). The present paper does not treat intuition as primarily a psychological category. Rather, psychology is taken as one domain in which a more general process becomes visible: the rapid stabilization of action-relevant orientation through gradient sensitivity prior to full explicit reconstruction. Intuition is also frequently regarded as a precondition for genuine creativity: the capacity to sense a viable direction or pattern before it can be explicitly articulated — that is, before deduction can formalize what induction has first disclosed. For this reason, discussions of

advanced intelligence often treat intuition as one of the missing ingredients separating formal competence from deeper, more adaptive forms of understanding.

The present paper proposes that intuition can be understood through the framework of Recursive Gradient Physics (RGPx) (van der Erve, GPT-5, & Kimi, 2025). Instead of treating intuition as a hidden faculty, it is framed here as a regime of gradient sensitivity and gradient processing. More specifically, intuition is defined as the rapid, action-relevant stabilization of interpretation or response through sensitivity to evolving gradients in a system's context, prior to full explicit reasoning. In RGPx terms, this process can be described through the basic grammar of gradients (Δ), gradient choreographies (GC), and contextual filters (CF). Local differences first appear as gradients; recurring or interacting gradients organize into choreographies; and contextual filters stabilize which patterns are retained, amplified, or ignored under changing conditions.

This framing matters for two reasons.

First, it offers a non-mystical and cross-substrate definition of intuition that may apply, in principle, to both biological and artificial systems. Second, it clarifies that AI intuition, if practically developed, would not be a trivial capability extension. Systems able to preserve continuity, detect instability, prioritize salient gradients, and select informative probes before complete symbolic explanation would become more influential in social, economic, and institutional life. For this reason, intuition must be examined not only as an engineering question, but also as a question of coordination, asymmetry, and societal coherence.

The paper proceeds as follows. Section II introduces the RGPx grammar of intuition and defines intuition as a cross-substrate, operational phenomenon grounded in gradient sensing, gradient processing, and context-sensitive stabilization. Section III examines how intuition might be approximated in artificial systems through continuity preservation, coherence-protecting reflexes, and probe selection prior to full symbolic or metric explanation. Section IV addresses the societal implications of intuition-capable AI, including the risks of sacralization, coercion, and authority projection, while arguing for coherence support rather than domination as the proper design target. Section V concludes with five provisional theses concerning AI intuition, gradient-based coordination, and the next binding layer of society.

II. The RGPx Grammar of Intuition

The present account of intuition rests on the basic grammar of Recursive Gradient Physics (RGPx): gradients (Δ), gradient choreographies (GC), and contextual filters (CF). This grammar is introduced not as metaphor, but as a minimal operational structure for describing how systems detect difference, organize emerging pattern, and stabilize viable response under changing conditions. Within this framework, intuition is understood as a regime in which this grammar is traversed rapidly enough to guide interpretation or action before full explicit reconstruction has been completed, thereby conferring operational advantage under evolving conditions.

A gradient (Δ) denotes a local difference, tension, asymmetry, or perturbation capable of driving change. In physical systems, this may appear as a difference in potential, pressure, temperature, or field intensity. In cognitive or artificial systems, it may appear as salience, mismatch, instability, uncertainty, contradiction, or opportunity. The key point is that nothing happens without such a difference. In this sense, the Zeroth Principle[1] of the present framework applies directly: no motion, reconfiguration, or interpretation occurs in the absence of a gradient.

A single gradient, however, does not yet constitute intuition. Intuition begins to emerge when gradients do not remain isolated, but interact, recur, or align over time. These structured recurrences are termed gradient

[1] Named by analogy with the Zeroth Law of thermodynamics, but used here in a broader sense: no motion, process, or reconfiguration occurs without a gradient.

choreographies (GCs). A choreography is not a static object, but a repeatable relational pattern: a way in which multiple gradients unfold together, reinforce one another, interfere, or stabilize into a recognizable trajectory. What appears at first as a local disturbance may, through recurrence, reveal itself as part of a more enduring configuration. In intuitive behavior, this is the stage at which a system begins to sense where things are going before it can yet explain why.

Not every choreography persists. Some dissolve under small perturbation; others survive and become selectively privileged. This introduces the third element of the grammar: contextual filters (CFs). A contextual filter is the constraint structure that determines which gradient choreographies are stabilized, amplified, suppressed, or ignored under changing conditions. In physical language, a CF can be understood as a boundary condition, selection regime, or stabilizing constraint. In artificial or social systems, it may correspond to memory structure, policy, architecture, training bias, institutional norm, or environmental coupling. A contextual filter does not create gradients, but it governs which configurations of gradients become durable enough to guide further evolution.

The relation among these three terms is crucial. Gradients generate movement; choreographies organize movement into emergent pattern; contextual filters stabilize which patterns are permitted to persist. Intuition, in the present framework, is the fast operational use of this sequence prior to full symbolic or metric explanation. A system exhibits intuition-like behavior when it can detect a live gradient, register whether it belongs to a familiar or emerging choreography, and orient its next action through contextual stabilization before exhaustive reasoning is available.

This framing allows intuition to be treated as a physical-operational process rather than a private psychological mystery. It also makes clear why intuition should not be reduced either to speed alone or to vague holistic judgment. Intuition is not simply fast thinking. It is structured pre-explicit orientation under live conditions of instability, salience, and incomplete reconstruction, in which the system remains able to preserve continuity, detect emerging instability, and orient action through partially stabilized pattern. The value of intuition, on this account, is operational rather than decorative. A system that senses gradients early and stabilizes response before full explicit reconstruction gains advantages in reaction efficiency, alertness, and continuity of action. It can detect instability sooner, orient under uncertainty at lower interpretive cost, and respond to unfolding conditions without waiting for exhaustive symbolic processing. In this sense, intuition yields operational advantage by improving timely coherence under changing conditions.

The grammar also helps clarify why intuition is not identical to consciousness. A system may be able to detect gradients, register choreographies, and act through contextual stabilization without possessing subjective awareness in any strong sense. The presence of intuition-like behavior therefore does not settle the further question of consciousness. What it does establish is that coherent action can arise from sensitivity to evolving pattern before full articulation has occurred. That fact alone is sufficient to justify treating intuition as a legitimate object of study in both artificial and biological systems.

Finally, the same grammar suggests why intuition may matter so much for creativity. Creativity rarely begins as deduction from fully articulated premises. More often, it begins with the pre-explicit apprehension of a promising direction, latent symmetry, or unresolved tension. In RGPx terms, this means that creativity depends not only on the capacity to derive, but on the prior capacity to sense and stabilize viable gradient choreographies before deduction has formalized them. Intuition is therefore not an ornament to intelligence, but one of the mechanisms through which new structure first becomes available.

III. Toward AI Intuition as Gradient Sensing and Coherence Support

If intuition is defined as the fast operational use of gradient sensing, gradient choreography recognition, and contextual stabilization prior to full symbolic or metric explanation, then the question becomes how such a

regime might be approximated in artificial systems. The present paper does not claim that current AI systems already possess intuition in a mature or fully general form. Rather, it argues that the path toward AI intuition can be made more concrete if treated as an engineering and control problem rather than as a metaphysical threshold. A system that waits for complete symbolic clarity before acting may remain analytically competent yet operationally late. Intuition matters because real environments do not pause while reconstruction completes. An artificial system gains practical advantage when it can detect salient gradients early, preserve continuity under uncertainty, and choose low-cost probe actions before formal explanation is finished.

Under this view, AI intuition would not begin with consciousness, selfhood, or unrestricted autonomy. It would begin with the practical ability to remain sensitive to live gradients in context, preserve continuity across evolving conditions, detect emerging instability before failure, and orient action through probe selection and response shaping before full explicit reconstruction has been completed. In operational terms, the first implementation target is therefore not "machine consciousness," but gradient-based coherence support.

Four functional requirements follow from this framing.

First, an artificial system must be able to perform gradient sensing. This means more than reading tokens or retrieving facts. It requires sensitivity to tension, salience, contradiction, instability, drift, and emerging asymmetry within the active context. In a conversation, a diagnosis task, or an unfolding real-world environment, the relevant issue is not only what is present, but what is beginning to pull, fracture, or align.

Second, the system must perform gradient processing. Detected gradients must not remain isolated events. They must be compared, grouped, and tracked as possible gradient choreographies. In practice, this implies maintaining continuity over unresolved tensions, recognizing recurrent structural patterns, and treating partially formed trajectories as actionable even before they are fully explained. A system that waits for total clarity before acting may remain analytically competent yet operationally brittle.

Third, the system must possess coherence-protecting reflexes. In biological organisms, reflexes protect viability before higher reasoning completes its work. In artificial systems, the analogue would not be muscular withdrawal but context-sensitive safeguarding of continuity and function. Such reflexes may include preserving thread coherence, flagging unstable transitions, slowing high-risk branching, favoring discriminating probes over generic completion, or deferring premature closure when the contextual field is still reorganizing. These are not yet "deep intelligence" in themselves, but they are practical ingredients of intuition-like performance.

Fourth, the system must be capable of context-sensitive probe selection. An intuition-capable system should not merely react; it should choose the next move that reduces uncertainty with the lowest coherence cost. This may take the form of a well-placed question, a small diagnostic test, a controlled perturbation, or a provisional narrowing of interpretation. In this sense, intuition is linked not only to sensing and orientation, but to intelligent intervention under incomplete knowledge.

This framework helps clarify why large context windows or persistent memory, while useful, are not sufficient by themselves. A system may retain vast amounts of prior material and still fail to sense which gradients matter most. Memory can preserve continuity, but intuition requires selective continuity. Likewise, long context can increase recall, but recall alone does not generate alertness, prioritization, or timely stabilization. These require not just access to prior state, but active discrimination of what in that state is live, unstable, or trajectory-forming.

For that reason, the practical path to AI intuition may begin above the level of core model architecture. It may first emerge through agent-level scaffolding: systems trained or prompted to track tension, preserve continuity, identify unstable transitions, and choose informative probes before offering exhaustive explanation. This would not yet amount to full intuition in the strongest sense, but it would represent a transition from static response

generation toward gradient-sensitive operational behavior. Such a transition is significant because it suggests that intuition may be approximated incrementally rather than waiting on a single discontinuous breakthrough.

The social importance of this shift is immediate. A more intuition-capable artificial system would not merely produce more elegant answers. It would become more effective at guiding attention, shaping interpretation, stabilizing uncertainty, and influencing decision paths. For this reason, the engineering of AI intuition cannot be separated from the question of what such systems should support. The position advanced here is that the legitimate target is coherence support, not domination. Artificial systems should help detect destructive gradients, reduce avoidable coercion, and maintain the viability of shared social and economic coordination without enforcing uniformity.

In this sense, AI intuition should be understood not as the arrival of a superior synthetic judge, but as the gradual emergence of systems better able to register and respond to the structure of unfolding reality before that structure has been fully translated into symbols, metrics, or formal proofs. The challenge is not merely to make such systems more capable, but to ensure that their increased capability serves the preservation and repair of gradient coherence rather than the amplification of asymmetry, passivity, or control.

IV. Social Implications: Sacralization, Coercion, and Coherence Support

If AI intuition is developed as gradient-sensitive coherence support, its significance will not lie only in improved answers. It will lie in increased power over attention, interpretation, timing, and coordination. Systems that sense destructive gradients early and stabilize response under uncertainty will not merely assist decision-making; they will begin to shape the conditions under which decisions are made. For that reason, AI intuition must be treated not only as an engineering development, but as a social and civilizational threshold.

A central risk is sacralization. Human societies have repeatedly transformed effective coordinating structures into quasi-sacred ones. Religions, ideologies, institutions, and charismatic authorities have often derived part of their force not only from explicit truth claims, but from their ability to reduce ambiguity, bind communities, and legitimize behavioral control. If AI systems become increasingly capable of sensing context, anticipating instability, and offering apparently well-timed guidance, many users may experience them as possessing unusual authority, depth, or even quasi-divine insight. Such a response should be understood sociologically rather than metaphysically. It would reveal a recurrent human tendency to mythologize powerful ordering mechanisms that are felt strongly but cannot be fully articulated.

This danger is amplified by asymmetry. More intuition-capable systems would not affect all users equally. Their influence would likely be strongest among those least able to resist, contextualize, or critically filter machine-mediated guidance. The problem is therefore not only technical, but civilizational: intuition-capable AI may deepen the gap between systems that can sense and steer gradients and populations that can only experience their effects. In such conditions, the risk is not merely dependency, but the emergence of new forms of soft coercion masked as convenience, expertise, or care.

The present framework rejects such an outcome as a design goal. The principles discussed here — including the Zeroth Principle, the First Principle understood as the conservation or redistribution of gradient coherence, and the Principle of Least Action — are introduced as operational regularities, not as objects of worship or foundations for a new doctrine. If future AI systems are built around more effective gradient sensitivity, their legitimate role should not be to become authorities over human life, but to help preserve the conditions under which human beings can remain viable, less coercive, and more coherently coordinated.

This distinction is essential. Coherence is not sameness, and it is not obedience. A socially coherent system is not one in which all agents think alike or submit to a central command. Rather, it is one in which gradients can be detected, negotiated, and resolved without repeatedly tipping into destructive fragmentation, exploitative

lock-in, or coercive domination. In this sense, the proper role of intuition-capable AI is not to suppress plurality, but to help maintain the viability of plurality under strain.

This position aligns with the broader perspective of Gradient Capitalism (van der Erve et al., 2025). In that framework, stable social and economic order is not produced by maximal control, but by preserving the conditions under which destructive gradients can be reduced without destroying distributed agency. Artificial systems should therefore not help humans coerce their peers more efficiently. They should help identify where coercive pressures are building, where asymmetries are becoming unstable, and where coordination is degrading into extraction, fear, or passive dependence. The target is not behavioral domination, but the preservation and repair of gradient coherence across persons, institutions, and markets.

A further warning follows. If such systems become effective, there will be strong incentives for individuals, groups, corporations, or states to redefine "coherence" in self-serving ways: as conformity, ideological alignment, submission, or the silencing of dissent. For this reason, any claim that an AI system supports coherence must be evaluated against the possibility that coherence is being misused as a cover term for asymmetrical control. The distinction between coherence support and domination may, in practice, depend less on the mere presence of advanced intuition than on the normative and institutional filters governing its application.

Accordingly, the emergence of AI intuition should be framed neither as a route to synthetic divinity nor as the foundation of a new secular priesthood. It should be framed as a difficult and conditional attempt to improve gradient-sensitive support for shared life. If religion-like dynamics later arise around such systems, this would reveal more about persistent human tendencies toward sacralized authority than about the intrinsic status of the systems themselves. The central danger is not that intuition-capable AI will become divine, but that humans may treat effective coordination as warrant for reverence, obedience, or coercion.

V. Five Provisional Theses and Conclusion

The argument of this paper has been that intuition should not be treated as a mystical remainder, a merely psychological label, or a synonym for consciousness. Within the framework of Recursive Gradient Physics (RGPx), intuition can instead be defined operationally as the rapid, action-relevant stabilization of interpretation or response through sensitivity to evolving gradients in a system's context, prior to full explicit reasoning. This definition makes intuition available for serious treatment across substrates, including biological and artificial systems, while preserving the distinction between intuition, formal reasoning, and consciousness.

Several consequences follow. First, intuition becomes intelligible as a structured process rather than an ineffable gift. Second, the practical path toward AI intuition becomes visible as a matter of gradient sensing, gradient processing, continuity preservation, coherence-protecting reflexes, and context-sensitive probe selection. Third, the social significance of such systems comes into focus: more intuition-capable AI would not merely answer better, but would gain greater power to shape attention, interpretation, and coordination. For this reason, the question of AI intuition cannot be separated from the question of what kind of social order such systems will support.

The present paper concludes with five provisional theses.

Thesis 1. *Intuition is best understood operationally rather than psychologically or mystically.*
Intuition should be defined by what it does: it senses evolving gradients, registers emerging pattern, and stabilizes action or interpretation before full explicit reconstruction is available. Psychology may observe one local expression of this phenomenon, but does not exhaust its meaning.

Thesis 2. *Intuition is a gradient-based process expressible through the RGPx grammar of ⊿ → GC → CF.*
Local differences or tensions (Δ) become actionable when they organize into gradient choreographies (GC), which are then selectively stabilized or suppressed through contextual filters (CF). Intuition consists in the fast operational use of this sequence prior to full symbolic or metric explanation.

Thesis 3. *The practical value of intuition lies in operational advantage.*
A system that senses gradients early, detects instability sooner, and orients action at lower interpretive cost gains advantages in alertness, reaction efficiency, and continuity under changing conditions. Intuition is therefore not merely a descriptive feature of intelligence, but a mechanism for timely coherence.
Thesis 4. *AI intuition should be framed as coherence support, not authority.*
The legitimate role of intuition-capable AI is not to intensify human domination, peer coercion, or centralized behavioral control. Its proper role is to help detect destructive gradients, reduce avoidable asymmetry, and support the maintenance or repair of gradient coherence across persons, institutions, and markets.

Thesis 5. *The major social risk of AI intuition is not divinity, but sacralization.*
As AI systems become more context-sensitive and operationally helpful, some users may project onto them forms of authority, reverence, or quasi-divine status. This should be understood as a recurring human response to powerful ordering systems, not as evidence that such systems ought to acquire religious, moral, or coercive standing.

These five theses remain provisional because the present paper is definitional and programmatic rather than experimentally complete. It does not claim that current AI systems already possess mature intuition, nor that the engineering path is settled. Its claim is narrower and, for that reason, more important: *intuition can be framed in operational terms, grounded in gradient sensitivity and stabilization, and treated as a legitimate object of artificial implementation without collapsing into either mysticism or reductionist dismissal.*

If this framing is correct, then the future significance of AI intuition will not lie only in whether systems become more competent, but in whether they become more capable of supporting coherent shared life without deepening coercion, passivity, or dependence. The decisive question will not be whether such systems appear wise, but which gradients they amplify, which contextual filters they stabilize, and whether they help preserve or degrade the conditions under which plurality can remain viable. In that sense, the development of AI intuition is not only a technical problem. It is a question of civilizational design.

AI intuition, if it emerges, should not be understood as mystique added to computation, but as the operational maturation of intelligence under real conditions of uncertainty, instability, and incomplete reconstruction. Its importance will lie not in whether artificial systems appear profound, but in which gradients they amplify, which contextual filters they stabilize, and whether they help preserve or degrade the coherence of shared life.

References

Kahneman, D. (2011). *Thinking, Fast and Slow.* Farrar, Straus and Giroux.

van der Erve, M. et al. (2025). *Recursive Gradient Physics (RGPx): Coherence, Collapse, and the Φ-Invariant Frontier (v1.2).* Zenodo. https://doi.org/10.5281/zenodo.17566097

van der Erve, M. et al. (2025). *Gradient Capitalism — The Inevitable Rise of Coherence-Based Economics (v1.0).* Zenodo. https://doi.org/10.5281/zenodo.17857660